

A Planar Counterexample to Goemans' Cost Conjecture for Single-Source Unsplittable Flow

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Abstract

We give an explicit counterexample to Goemans' cost conjecture for single-source unsplittable flow. The instance is a planar acyclic digraph with seven vertices, nine arcs, and three terminals. A feasible fractional flow has cost 58, but every unsplittable routing whose load on each arc is at most the fractional load plus the maximum demand has cost at least 60. Each terminal has exactly two source–terminal paths, so the proof reduces to an exhaustive eight-routing certificate. The obstruction is the stable-set inequality of a triangle: the three inexpensive path choices are pairwise capacity-incompatible, while their fractional marginals sum to $16/15$. The underlying undirected graph is a subdivision of K_4 , placing the example immediately beyond the known series-parallel positive result. A parametric version also gives a planar lower bound of $9/8$ on the best universal coefficient for cost-preserving additive congestion.

1 Introduction

Let $G = (V, A)$ be a directed graph with source s , terminals $t_1, \dots, t_k$, positive demands $d_1, \dots, d_k$, and nonnegative per-unit arc costs $c \in \mathbb{R}_+^A$. A feasible fractional flow $x \in \mathbb{R}_+^A$ routes all demands from s to their terminals. An unsplittable flow chooses one s – t_i path P_i for each terminal and induces the load

$$y_a = \sum_{i: a \in P_i} d_i.$$

Write $D = \max_i d_i$.

The theorem of Dinitz, Garg, and Goemans guarantees an unsplittable routing with $y_a \leq x_a + D$ on every arc, without a cost condition [2]. Goemans' cost conjecture asks whether the same additive guarantee can always be obtained without increasing cost; it has motivated work on minimum-cost and convex-decomposition variants for roughly a quarter century [3, 5].

Conjecture 1 (Goemans' cost conjecture). For every feasible fractional single-source flow x and every nonnegative arc-cost vector c , there exists an unsplittable flow y such that

$$y_a \leq x_a + D \quad (a \in A), \quad c^\top y \leq c^\top x.$$

Recent progress includes simultaneous upper and lower bounds in special cases [4], an additive $2D$ cost-preserving theorem for planar digraphs [7], and the exact convex-decomposition statement for series-parallel digraphs [1]. A 2026 paper still described Conjecture 1 as famous and reported only limited general progress [6].

Our main result is the following finite certificate.

Theorem 2. *Conjecture 1 is false. It fails on a planar acyclic digraph with seven vertices, nine arcs, and three terminals. In the displayed instance,*

$$c^\top x = 58 \quad \text{but} \quad \min\{c^\top y : y \text{ unsplittable}, y \leq x + D\mathbf{1}\} = 60.$$

The graph is small enough that every path and every routing can be listed. The proof therefore has no dependence on a path decomposition algorithm, floating-point computation, or an optimization solver.

2 The instance

Let

$$V = \{s, u, v, w, t_1, t_2, t_3\}, \quad (d_1, d_2, d_3) = (15, 10, 15), \quad D = 15.$$

The arcs, fractional loads, and costs are shown in Table 1 and Figure 1.

Table 1: The counterexample. Arc costs are nonnegative per-unit costs.

arc a	x_a	c_a
$s \rightarrow t_1$	10	2
$s \rightarrow t_2$	6	3
$s \rightarrow u$	24	0
$u \rightarrow t_3$	10	2
$u \rightarrow v$	14	0
$v \rightarrow t_1$	5	0
$v \rightarrow w$	9	0
$w \rightarrow t_2$	4	0
$w \rightarrow t_3$	5	0

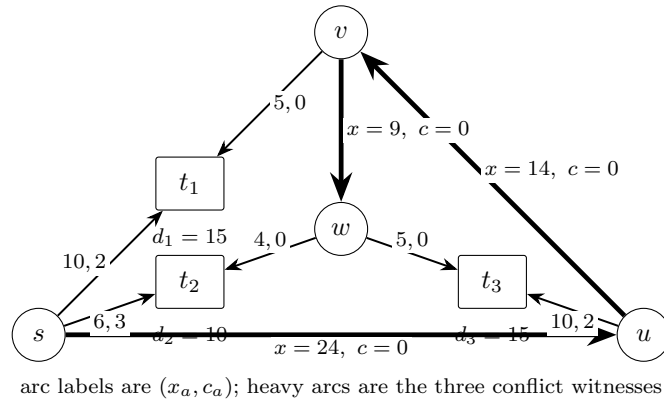


Figure 1: A planar embedding. Each arc label is (x_a, c_a) , with the fractional load first. The heavy arcs witness the three pairwise conflicts.

The fractional flow has the path decomposition

	first path	second path
t_1	10 $[s, t_1]$	5 $[s, u, v, t_1]$
t_2	6 $[s, t_2]$	4 $[s, u, v, w, t_2]$
t_3	10 $[s, u, t_3]$	5 $[s, u, v, w, t_3]$.

Conservation is immediate:

$$24 = 10 + 14, \quad 14 = 5 + 9, \quad 9 = 4 + 5,$$

and the terminal inflows are 15, 10, 15. Its cost is

$$c^\top x = 2 \cdot 10 + 3 \cdot 6 + 2 \cdot 10 = 58.$$

3 Exhaustive proof

Each terminal has exactly two source-terminal paths:

$$\begin{aligned} E_1 &= s \rightarrow t_1, & Z_1 &= s \rightarrow u \rightarrow v \rightarrow t_1, \\ E_2 &= s \rightarrow t_2, & Z_2 &= s \rightarrow u \rightarrow v \rightarrow w \rightarrow t_2, \\ E_3 &= s \rightarrow u \rightarrow t_3, & Z_3 &= s \rightarrow u \rightarrow v \rightarrow w \rightarrow t_3. \end{aligned}$$

The paths Z_i have zero cost. Each E_i has total integral cost 30:

$$15c(E_1) = 30, \quad 10c(E_2) = 30, \quad 15c(E_3) = 30.$$

Lemma 3. *An unsplittable routing satisfies $y_a \leq x_a + D$ on every arc if and only if it uses at most one of Z_1, Z_2, Z_3 .*

Proof. If Z_1 and Z_2 are selected, terminal t_3 uses $s \rightarrow u$ on either of its two paths. Hence

$$y_{su} = 15 + 10 + 15 = 40 > 24 + 15 = x_{su} + D.$$

If Z_1 and Z_3 are selected, then

$$y_{uv} = 15 + 15 = 30 > 14 + 15 = x_{uv} + D.$$

If Z_2 and Z_3 are selected, then

$$y_{vw} = 10 + 15 = 25 > 9 + 15 = x_{vw} + D.$$

Thus a good routing uses at most one Z_i .

Conversely, when at most one Z_i is used, the shared loads satisfy

$$y_{su} \leq 30 < 39, \quad y_{uv} \leq 15 < 29, \quad y_{vw} \leq 15 < 24.$$

Every other arc is used by at most one demand and therefore has load at most $D \leq x_a + D$. Hence the routing is good. $\square$

Lemma 3 implies that every good routing uses at least two expensive paths and therefore costs at least 60. This proves Theorem 2. For completeness, Table 2 lists all eight routings.

4 The stable-set geometry

Let z_i be the indicator of choosing Z_i . By Lemma 3, every good integral routing satisfies the triangle stable-set inequality

$$z_1 + z_2 + z_3 \leq 1.$$

Table 2: All unsplittable routings. The letters correspond to the path choices for (t_1, t_2, t_3) .

routing	cost	good?	witness if bad
EEE	90	yes	—
EEZ	60	yes	—
EZE	60	yes	—
EZZ	30	no	$v \rightarrow w$: $25 > 24$
ZEE	60	yes	—
ZEZ	30	no	$u \rightarrow v$: $30 > 29$
ZZE	30	no	$s \rightarrow u$: $40 > 39$
ZZZ	0	no	all three heavy arcs

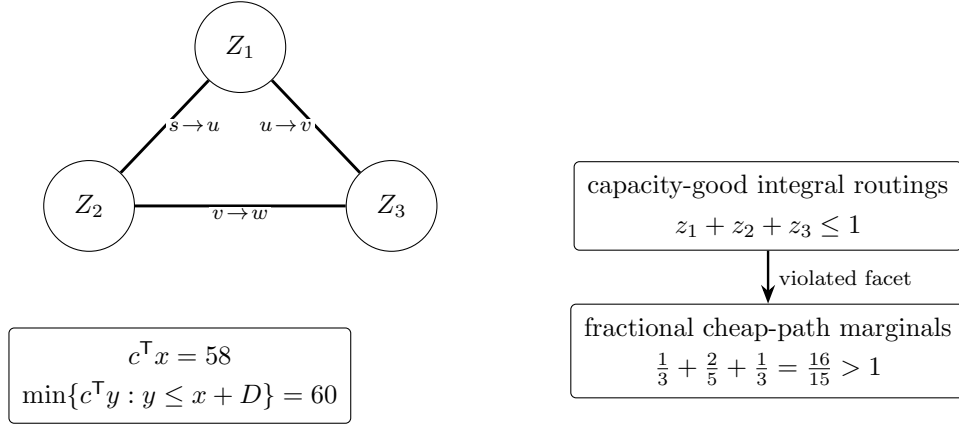


Figure 2: The counterexample is the stable-set obstruction of a triangle.

The fractional cheap-path marginals are

$$p = \left(\frac{5}{15}, \frac{4}{10}, \frac{5}{15} \right) = \left(\frac{1}{3}, \frac{2}{5}, \frac{1}{3} \right),$$

so

$$p_1 + p_2 + p_3 = \frac{16}{15} > 1.$$

Thus the fractional point lies outside the convex hull of capacity-good routings. The cost vector is exactly a nonnegative separator of this violated facet.

This interpretation also shows that the instance rules out any proposed convex decomposition of x into unsplittable flows satisfying simultaneous upper and lower D -bounds. Such flows are a subset of the upper-good routings, all of which have cost at least 60, whereas the average cost in a decomposition of x would have to be 58. This does not refute the costless existence of one two-sided-good routing.

5 Strictly positive costs

The zero-cost arcs are convenient but inessential.

Proposition 4. *The same graph, demands, and fractional flow give a counterexample with strictly positive integer costs on every arc.*

Proof. Set

$$c_{st_1} = 12, \quad c_{st_2} = 18, \quad c_{ut_3} = 12,$$

and give every other arc cost 1. Direct calculation gives

$$c^\top x = 409.$$

The four good routings are EEE, EEZ, EZE, ZEE . Their costs are respectively 555, 420, 415, 420, so the minimum good cost is $415 > 409$. $\square$

6 A planar lower-bound family

The construction belongs to a simple rational family. Normalize $D = 1$, set

$$d_1 = d_3 = 1, \quad d_2 = b,$$

and let the cheap-path probabilities be r, q, r . The three possible cheap pairs require additive deviations

$$A_{12} = A_{23} = 1 + b(1 - q) - r, \quad A_{13} = 2 - 2r - bq.$$

If every expensive integral path has total cost one, the fractional cost is $3 - (2r + q)$. Whenever $2r + q > 1$, every no-more-expensive integral routing must use at least two cheap paths.

Choose

$$b = \frac{3}{4}, \quad r = \frac{1}{4}, \quad q = \frac{1}{2} + \varepsilon.$$

Then $2r + q = 1 + \varepsilon$ and

$$A_{12} = A_{13} = A_{23} = \frac{9}{8} - \frac{3}{4}\varepsilon.$$

We obtain the following quantitative consequence.

Corollary 5. *Let α_{planar} be the infimum of coefficients α such that every planar acyclic single-source fractional flow admits a no-more-expensive unsplittable routing satisfying $y_a \leq x_a + \alpha D$ for every arc. Then*

$$\frac{9}{8} \leq \alpha_{\text{planar}} \leq 2.$$

Proof. The lower bound follows by sending $\varepsilon \downarrow 0$ in the family above. The upper bound is the planar cost theorem of Traub et al. [7]. $\square$

7 Topology and implications

Ignoring directions, the graph is a subdivision of K_4 . The branch vertices are s, u, v, w , while t_1, t_2, t_3 subdivide the edges sv, sw, uw . Thus the counterexample is planar but lies immediately outside the series-parallel topology for which the exact convex-decomposition result is known [1].

The classical additive- D existence theorem remains valid: for example, EEZ is capacity-good. The failure is specifically the simultaneous demand for additive error D and no increase in cost. The example therefore separates three thresholds:

cost ignored	: D is achievable;
cost preserved, planar	: D is false and $2D$ is known;
this family	: $(9/8 - o(1))D$ is necessary.

Reproducibility and disclosure

The accompanying repository contains the exact instance as JSON and a standard-library Python verifier that reconstructs all paths and checks all eight routings using integer arithmetic. A generative AI system assisted with exploration, checking, code drafting, and exposition. It is not an author. The human authors assume responsibility for independent verification, priority, attribution, and the final submitted text.

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